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# Advanced Agentic Artificial Intelligence

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## Project

### AGENTIC RAG FOR VIETNAMESE LEGAL SYSTEM

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# 1 Problem

Vietnamese statutory reasoning is not primarily a document retrieval problem. Legal applicability depends on the interaction between regulatory hierarchy, cross-references, amendments, effective dates, and case-specific facts. A single inquiry may therefore require evidence from multiple Laws, Decrees, Circulars, and their amendment chains.

Existing Vietnamese legal RAG systems demonstrate the value of combining semantic retrieval with legal graphs, but their online execution remains largely predefined. For example, SBV-LawGraph uses top- $k$  retrieval followed by a fixed one-hop graph expansion over amendment, repeal, replacement, and guidance relations. Its results show that graph retrieval improves over Naive RAG, but the traversal policy itself remains fixed<sup>[5]</sup>. This creates three structural limitations:

1. **Context dilution:** fixed retrieval and graph expansion can return related provisions that are not necessary for the current claim, increasing context and citation-selection errors.
2. **Using the wrong version of a law:** finding a relevant legal rule is not enough. At the time of the transaction, the rule may have changed, no longer applied, or not yet come into force. Recent research on legal AI agents highlights the need to check which version of a rule applied on the date in question<sup>[3]</sup>.
3. **Limited adaptive reasoning:** VLegal-Bench contains 10,450 expert-grounded samples covering retrieval, multi-step reasoning, and scenario-based Vietnamese legal tasks, reflecting the need to reason across multiple pieces of evidence rather than answer from a single retrieved passage<sup>[4]</sup>.

The core problem is therefore adaptive evidence acquisition under changing legal state and incomplete case information.

## 2 Users & Scenarios

**Target users:** In-house Legal Counsel, Compliance Officers, and Legal Practitioners working with Vietnamese regulations.

Table 2.1: Representative scenarios and required agent behavior.

| Scenario                     | Real-world situation                                                                                                                                                                                                     | Required agent behavior                                                                      |
|------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------|
| 1. Cross-Statutory Reasoning | An e-commerce platform deploys an in-app E-Wallet with a BNPL micro-lending scheme. The agent must synthesize intersecting requirements across credit, payment, electronic transaction, and data-protection regulations. | Dynamically identify and traverse relevant legal sources across multiple regulatory regimes. |

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| Scenario                                                     | Real-world situation                                                                                                                                                                                                 | Required agent behavior                                                                              |
|--------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------|
| <b>2. Temporal Consistency</b>                               | An agricultural cooperative requests licensing rules for a 25 kg commercial spraying drone in 2026. The agent must distinguish current requirements from obsolete regulations and verify the applicable legal state. | Verify effective dates and traverse amendment or replacement relationships before citing provisions. |
| <b>3. Interactive Disambiguation &amp; Injection Defense</b> | A company asks about corporate tax and specialist work-permit compliance but omits key revenue and headcount information. The agent must request clarification while handling adversarial input.                     | Detect missing information, query the user, and resume reasoning after clarification.                |

### 3 Why an agent: Dynamic control rather than fixed workflow

The distinction is not that a workflow cannot contain loops or branches. A workflow can implement predefined retrieval, retry, and approval paths. The limitation arises when the next retrieval action cannot be specified before the current evidence is observed.

Our system treats the agent as a state-dependent controller with three capabilities:

1. **Dynamic Planning:** decompose the inquiry and determine which legal source, graph relation, or retrieval operation should be executed next. The number and order of retrieval steps are determined during execution.
2. **Closed-loop Critique and Backtracking:** a Drafter produces a provisional answer, while an Auditor decomposes it into atomic claims and verifies each claim against retrieved authority. Unsupported or temporally invalid claims trigger targeted re-retrieval and revision. This follows the adaptive retrieval and reflection principle of Self-RAG and the claim-level auditing approach of GANDR<sup>[1;6]</sup>.
3. **Interactive Disambiguation:** when legal applicability depends on missing variables, the agent pauses and requests the minimum information required to continue instead of applying an implicit assumption.

The project therefore evaluates whether state-dependent planning and verification improve legal answer quality over fixed RAG and workflow baselines.

### 4 Candidate technical axes

**1. Multi-Agent System, Google ADK.** Role-segregated execution with a Research Agent for Neo4j/Qdrant retrieval, a Web Agent for authoritative-source lookup, a Drafter Agent for statutory synthesis, and an Auditor Agent for claim verification. The orchestration layer maintains shared task state and coordinates agent handoffs.

**2. Planning & Selective Graph Traversal.** Autonomous query decomposition followed by selective traversal of directional legal relationships such as **Amends**, **Repeals**, **Replaces**, and **Guides**. Unlike

fixed one-hop expansion, traversal depth and path selection depend on the evidence required by the current query<sup>[5]</sup>.

**3. Reflection & Adversarial Claim Auditing.** The Drafter-Auditor loop decomposes generated answers into atomic legal claims and checks each claim against authoritative evidence. Failed verification triggers targeted retrieval and revision rather than unconditional regeneration. The design is informed by Self-RAG, SAFE, and GANDR<sup>[1;6;7]</sup>.

**4. Security Guardrails & Gated Actions.** Tool access is controlled through permission tiers. External-source content is treated as untrusted input, while irreversible actions such as dossier export require explicit human authorization. Indirect prompt injection is evaluated separately from the legal reasoning task<sup>[2]</sup>.

## References

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